# Kinetic Modeling of Cell Signaling

## Module 1: Reaction networks as dynamical systems

Jim Faeder

MSCBIO 2025 · Introduction to Bioinformatics Programming in Python

Fall 2026

# Resources

- **BioNetGen** software, documentation, and tutorials: bionetgen.org
- **VS Code extension**: search the Extensions panel for *BioNetGen*
- **BNG Playground**: write and simulate models in your browser, no installation ruleworld.github.io/bngplayground
- **bngsim**: fast ODE and stochastic simulation from Python bngsim.readthedocs.io · `pip install bngsim bionetgen`
- **Course repository**: models, notebooks, slides and handouts github.com/jrfaeder/RBM_Intro_BioNetGen



Course repository



BNG Playground

# The book

*Rule-Based Modeling of Biological Systems: Theory and practice with BioNetGen* is an open book that develops the material of this course in more depth, with every model, figure, and exercise.

ruleworld.github.io/bionetgen-book

For this module, read:

- Chapter 1: Reaction networks as dynamical systems
- Chapter 7: Analyzing models (extends the pulse-shape exercise)



The book

# Additional reading

- J. R. Faeder, M. L. Blinov, and W. S. Hlavacek (2009). **Rule-based modeling of biochemical systems with BioNetGen.** *Methods Mol. Biol.* **500**, 113–167. doi:10.1007/978-1-59745-525-1_5
  A thorough introduction to BNGL and the BioNetGen software.
- J. A. P. Sekar and J. R. Faeder (2012). **Rule-based modeling of signal transduction: a primer.** *Methods Mol. Biol.* **880**, 139–218. doi:10.1007/978-1-61779-833-7_9
  A self-contained tutorial: building a receptor model step by step, model assumptions, compartments, and a library of example rules.
- L. A. Harris et al. (2016). **BioNetGen 2.2: advances in rule-based modeling.** *Bioinformatics* **32**, 3366–3368. Free full text (PMC)
  What version 2.2 added: rate laws beyond mass action, faster stochastic and hybrid simulation, and SBML import.

The two *Methods in Molecular Biology* chapters are available through the Pitt library: follow the DOI link on campus or after signing in to the library.

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5. explain how a **negative feedback** loop shapes the dynamics of a signaling response

Part 1

# Why model reaction networks?

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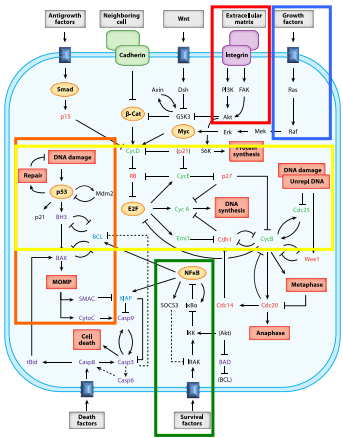
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## What a model gives us

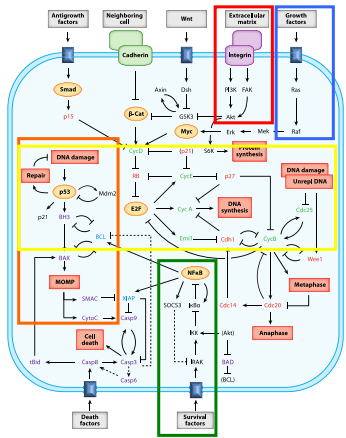
- A precise statement of a mechanism
- Predictions of dynamics we cannot easily measure
- A way to test whether a mechanism can explain data
- Hypotheses for the next experiment

# Toward an integrated model of the cell




Cell fate is decided by many interacting signaling modules: growth-factor signaling, the cell cycle, DNA damage, apoptosis.

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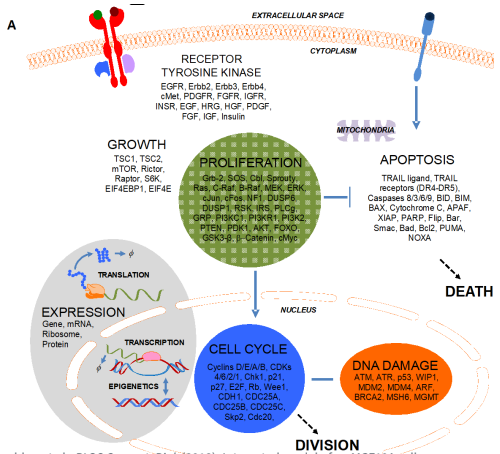



Tyson and Novak, *Annu. Rev. Phys. Chem.* **61**, 219–240 (2010)

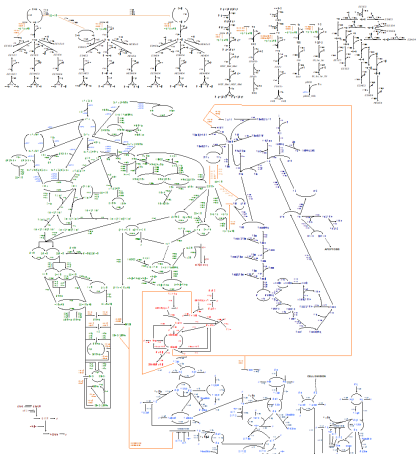
Cell fate is decided by many interacting signaling modules: growth-factor signaling, the cell cycle, DNA damage, apoptosis.

**How do we link these modules together** into models that predict how cells respond?

# Large models are built from simple reaction steps




Bouhaddou et al., *PLOS Comput. Biol.* (2018). Integrated model of an MCF10A cell.

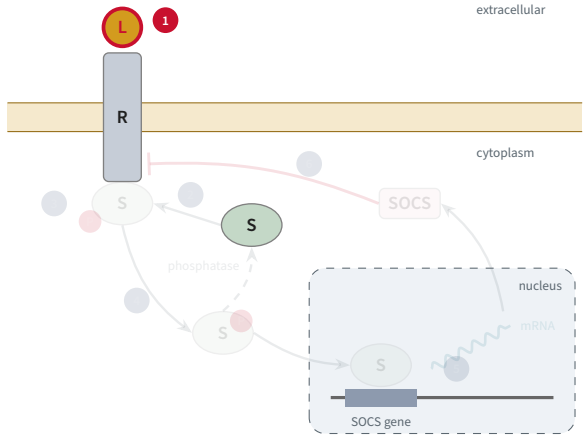


Part 2

# A cytokine signaling circuit

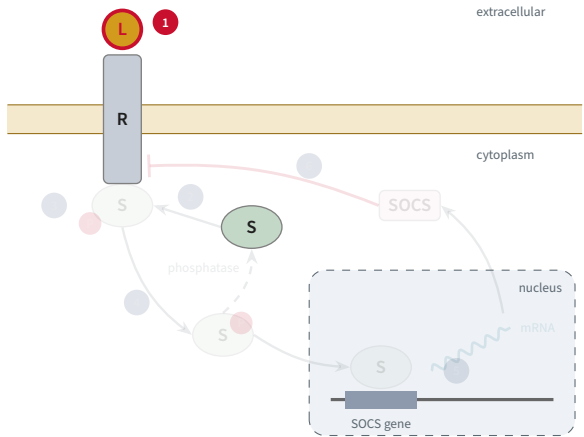
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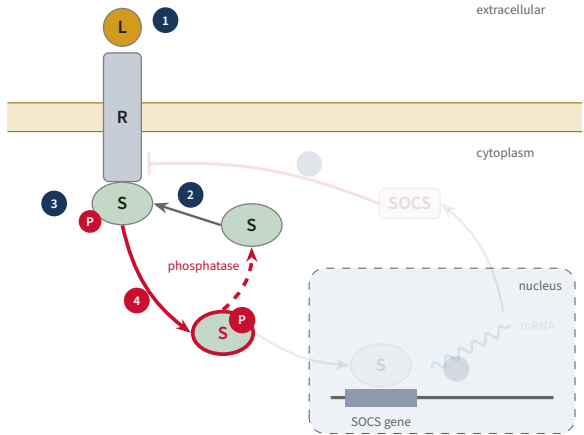
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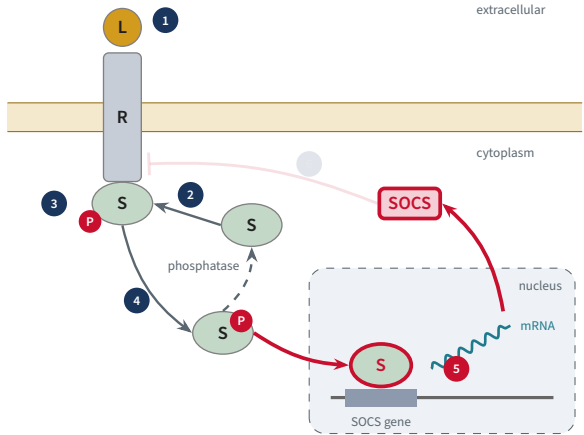
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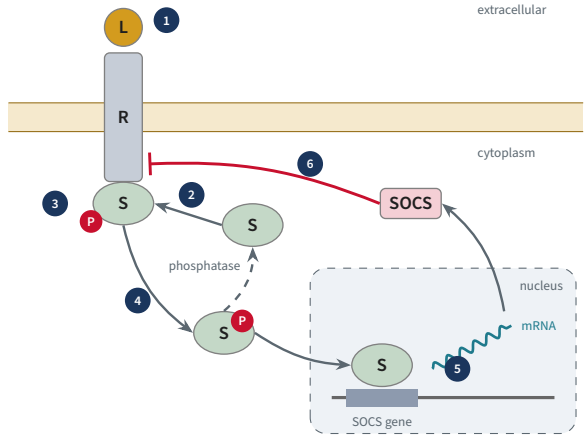
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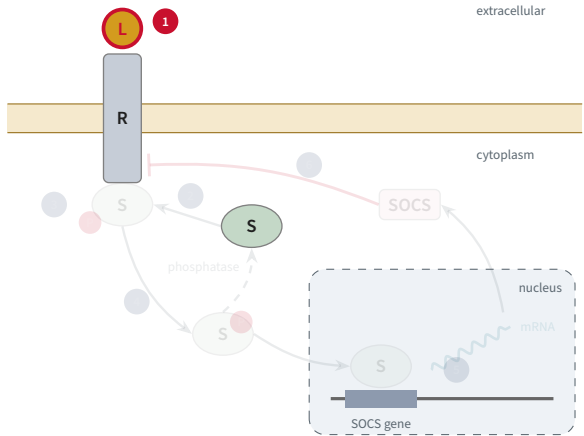
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- SOCS binds the receptor and **shuts off further STAT activation**: negative feedback.



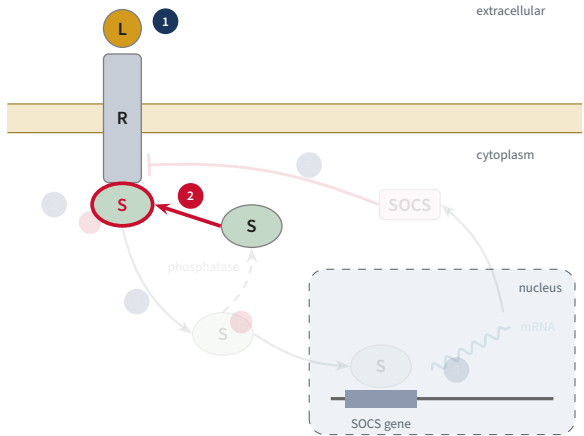
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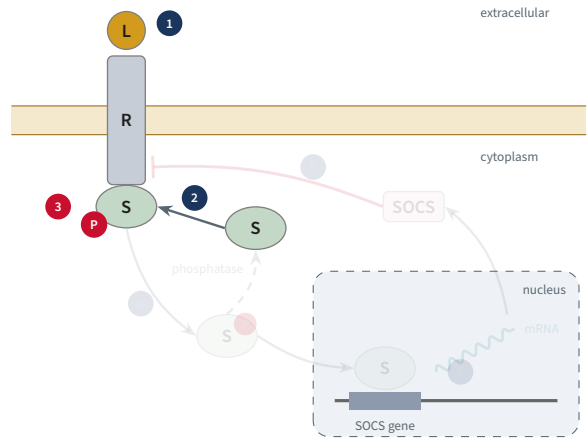
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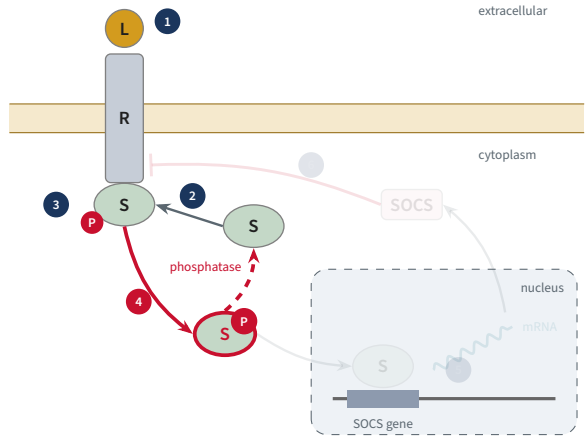
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3. Receptor-bound STAT is activated



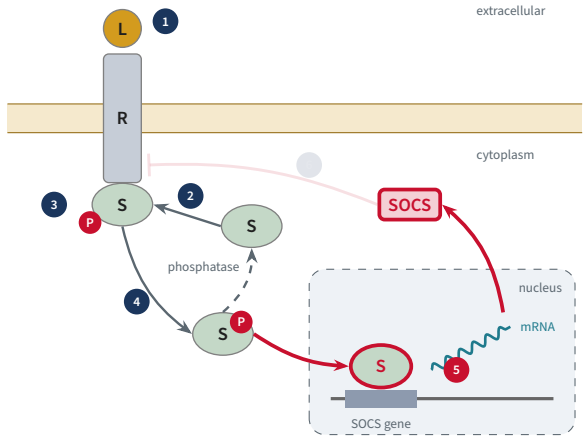
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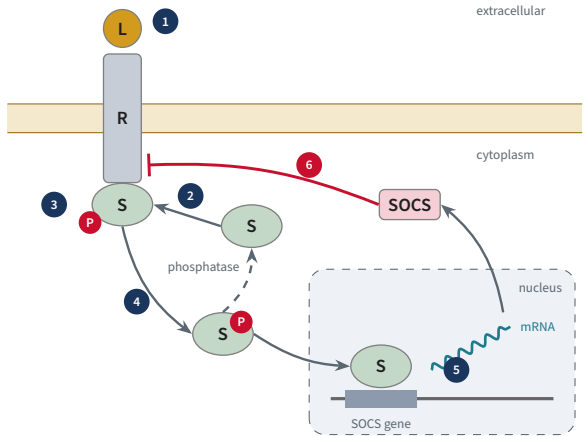
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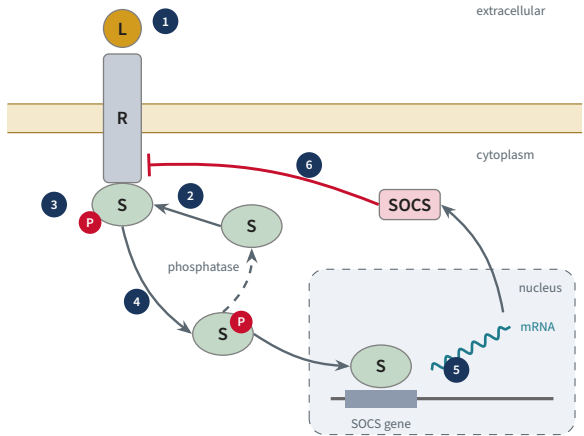
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Each step adds reactions to the previous model. The files are `models/stat_step1_binding.bngl` through `stat_step6_feedback.bngl`.

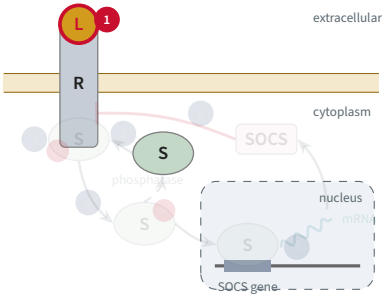


Part 3

# Step 1: Ligand–receptor binding

# Binding is a reversible pair of reactions

$$L + R \underset{k_{off}}{\overset{k_{on}}{\rightleftharpoons}} LR$$




Units: $k_{on}$ in $\mu M^{-1} min^{-1}$, $k_{off}$ in $min^{-1}$

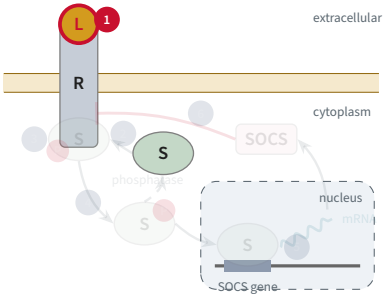
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Two elementary reactions, each with its own rate:

binding: $v_1 = k_{on}\,[L]\,[R]$

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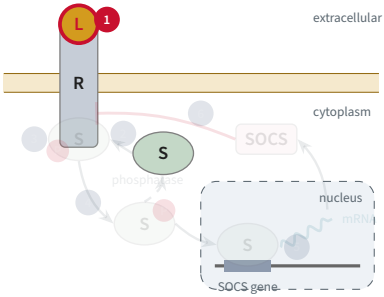
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## Law of mass action

The rate of an elementary reaction is proportional to the product of the concentrations of its reactants. The proportionality constant is the **rate constant**.




Units: $k_{on}$ in $\mu M^{-1} min^{-1}$, $k_{off}$ in $min^{-1}$

# Each reaction contributes to the rate of change of its species

| | $v_1$ (bind) | $v_2$ (unbind) |
|---|---|---|
| L | $-1$ | $+1$ |
| R | $-1$ | $+1$ |
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$$\frac{d[\mathrm{L}]}{dt} = -k_{\mathrm{on}}[\mathrm{L}][\mathrm{R}] + k_{\mathrm{off}}[\mathrm{LR}]$$

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**Ordinary differential equations (ODEs) follow mechanically from the list of reactions and their rate laws.**

# The stoichiometry matrix gives a compact general form

Collect the concentrations in a vector **X** and the reaction rates in **v**:

$$\frac{d\mathbf{X}}{dt} = \mathbf{S}\,\mathbf{v}(\mathbf{X}), \qquad \frac{d}{dt}\begin{pmatrix}[\mathrm{L}]\\ [\mathrm{R}]\\ [\mathrm{LR}]\end{pmatrix} = \underbrace{\begin{pmatrix}-1 & +1\\ -1 & +1\\ +1 & -1\end{pmatrix}}_{\mathbf{S}} \underbrace{\begin{pmatrix}k_{\mathrm{on}}[\mathrm{L}][\mathrm{R}]\\ k_{\mathrm{off}}[\mathrm{LR}]\end{pmatrix}}_{\mathbf{v}(\mathbf{X})}$$

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- **S** has one row per species and one column per reaction.
- This form holds for *any* reaction network; only **S** and **v** change.

# Conservation laws and equilibrium

**Conservation.** Binding only moves molecules between free and bound forms:

$$[\mathrm{L}] + [\mathrm{LR}] = L_0$$
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**Small models can be solved by hand. Real signaling networks cannot, so we solve the ODEs numerically.**

# ODEs for reaction networks are solved numerically

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- Reactions occur on very different time scales, which makes the system **stiff**: simple methods need impractically small steps.
- Production solvers (CVODE, LSODA) adapt the step size and use implicit methods for stiff systems.

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$$\mathbf{X}(t + \Delta t) \approx \mathbf{X}(t) + \Delta t\ \mathbf{S}\,\mathbf{v}(\mathbf{X}(t))$$

In practice:

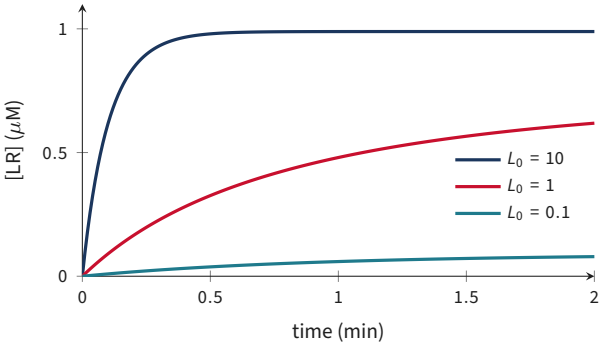
- The equations are **nonlinear** and usually have no closed-form solution.
- Reactions occur on very different time scales, which makes the system **stiff**: simple methods need impractically small steps.
- Production solvers (CVODE, LSODA) adapt the step size and use implicit methods for stiff systems.

## In BioNetGen

`simulate({method=>"ode",...})` uses CVODE.

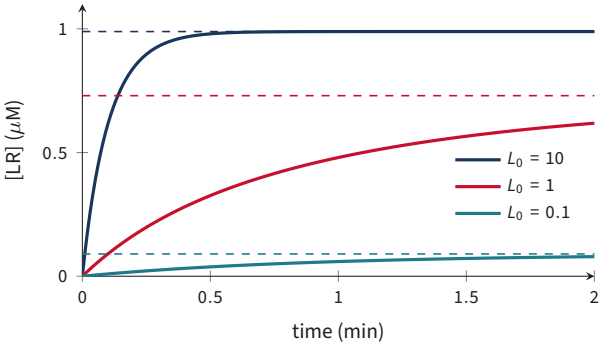
In Python, `bngsim.Simulator(model, method="ode")` runs the same model with CVODE.

# Binding approaches equilibrium faster and further at higher ligand




$R_0 = 1\,\mu\text{M}$, $k_{on} = 1\,\mu\text{M}^{-1}\text{min}^{-1}$,
$k_{off} = 0.1\,\text{min}^{-1}$ ($K_D = 0.1\,\mu\text{M}$).

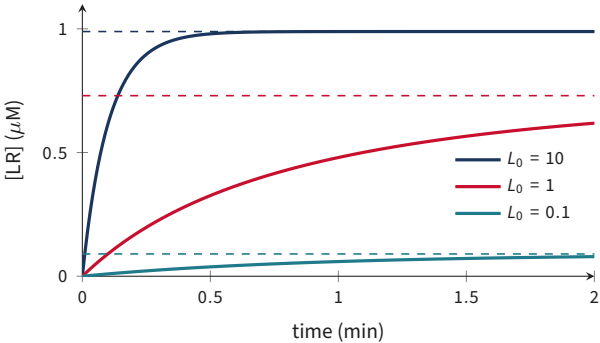
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Dashed lines: $[\text{LR}]_{eq}$ from the formula on the previous slide.

# Binding approaches equilibrium faster and further at higher ligand




$R_0 = 1\ \mu\text{M}$, $k_{\text{on}} = 1\ \mu\text{M}^{-1}\text{min}^{-1}$, $k_{\text{off}} = 0.1\ \text{min}^{-1}$ ($K_D = 0.1\ \mu\text{M}$).

Dashed lines: $[\text{LR}]_{\text{eq}}$ from the formula on the previous slide.

The relaxation rate is $k_{\text{on}}([\text{L}] + [\text{R}]) + k_{\text{off}}$ near equilibrium, so higher ligand also means a faster response.

# The same model in BNGL

```
begin model

begin parameters
  L0        10     # ligand (uM)
  R0        1      # receptor (uM)
  kon       1      # binding (1/uM/min)
  koff      0.1    # unbinding (1/min)
end parameters

begin molecule types
  L
  R
  LR
end molecule types

begin species
  L    L0
  R    R0
end species
```

# The same model in BNGL

```
begin model

begin parameters
  L0        10      # ligand (uM)
  R0        1       # receptor (uM)
  kon       1       # binding (1/uM/min)
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end parameters

begin molecule types
  L
  R
  LR
end molecule types

begin species
  L    L0
  R    R0
end species
```

```
begin observables
  Molecules Lfree   L
  Molecules Rfree   R
  Molecules LR      LR
end observables

begin reaction rules
  # Ligand binds receptor
  LRbind: L + R <-> LR kon, koff
end reaction rules

end model

generate_network({overwrite=>1})
simulate({method=>"ode",t_end=>2,n_steps=>200})
```

# The same model in BNGL

```
begin model

begin parameters
  L0        10      # ligand (uM)
  R0        1       # receptor (uM)
  kon       1       # binding (1/uM/min)
  koff      0.1     # unbinding (1/min)
end parameters

begin molecule types
  L
  R
  LR
end molecule types

begin species
  L    L0
  R    R0
end species
```

```
begin observables
  Molecules Lfree   L
  Molecules Rfree   R
  Molecules LR      LR
end observables

begin reaction rules
  # Ligand binds receptor
  LRbind: L + R <-> LR kon, koff
end reaction rules

end model

generate_network({overwrite=>1})
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```

- **observables**: quantities to output

# The same model in BNGL

```
begin model

begin parameters
  L0        10      # ligand (uM)
  R0        1       # receptor (uM)
  kon       1       # binding (1/uM/min)
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end parameters

begin molecule types
  L
  R
  LR
end molecule types

begin species
  L     L0
  R     R0
end species
```

```
begin observables
  Molecules Lfree   L
  Molecules Rfree   R
  Molecules LR      LR
end observables

begin reaction rules
  # Ligand binds receptor
  LRbind: L + R <-> LR kon, koff
end reaction rules

end model

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- **reaction rules**: <–> is reversible; forward and reverse rate constants

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begin model

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  L0        10      # ligand (uM)
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end parameters

begin molecule types
  L
  R
  LR
end molecule types

begin species
  L    L0
  R    R0
end species
```

```
begin observables
  Molecules Lfree   L
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  Molecules LR      LR
end observables

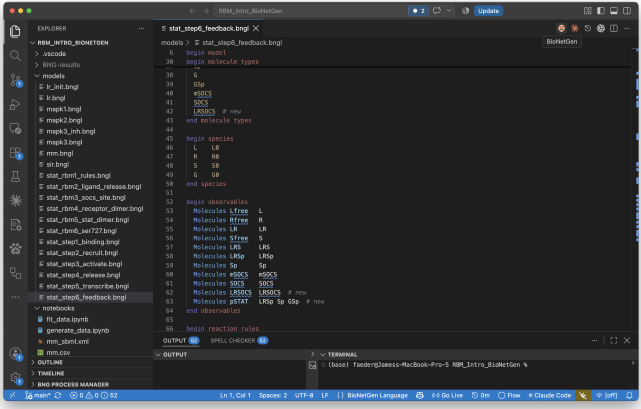
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  LRbind: L + R <-> LR kon, koff
end reaction rules

end model

generate_network({overwrite=>1})
simulate({method=>"ode",t_end=>2,n_steps=>200})
```

- **observables**: quantities to output
- **reaction rules**: `<->` is reversible; forward and reverse rate constants
- **actions** after `end model`: generate the network, then simulate

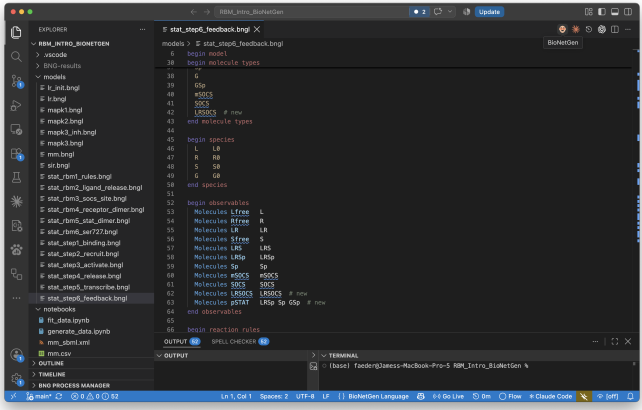
# Running a model in VS Code




- Open a `.bngl` file

VS Code extension: Ali Sinan Saglam

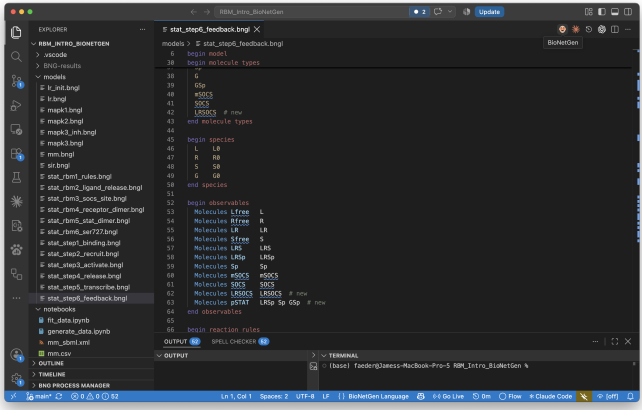
# Running a model in VS Code




VS Code extension: Ali Sinan Saglam

- Open a `.bngl` file
- Click the **run** button in the editor toolbar

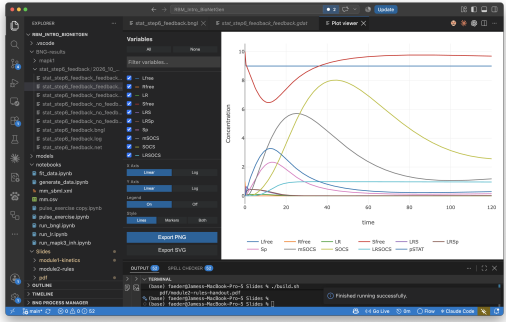
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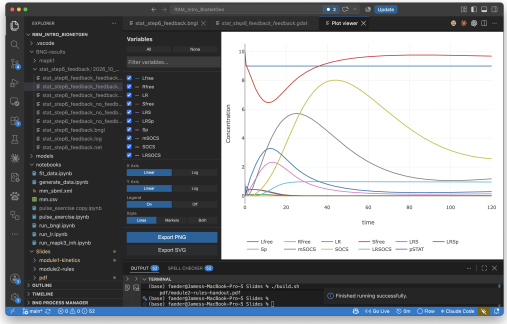
- Open a `.bngl` file
- Click the **run** button in the editor toolbar
- Results go to `BNG-results/<model>/<timestamp>/`

# Interactive plotting and Python notebooks

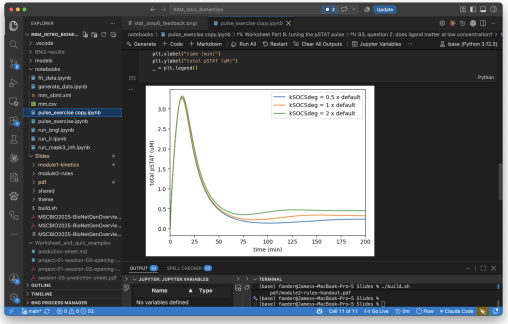



Built-in plotting of `.gdat` output

# Interactive plotting and Python notebooks

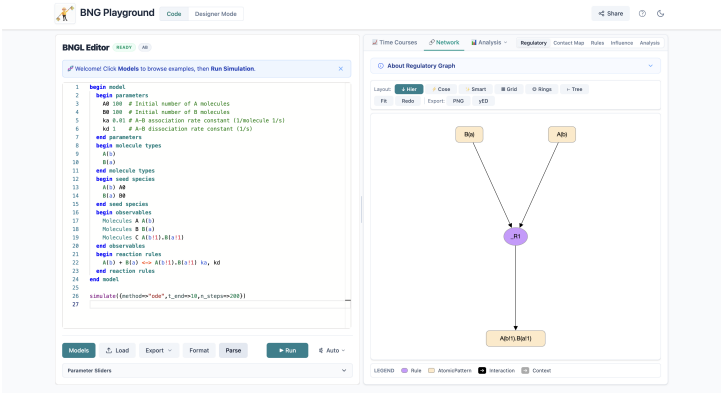



Built-in plotting of `.gdat` output




Deeper analysis in Jupyter with bngsim

# BNG Playground: BioNetGen in your browser

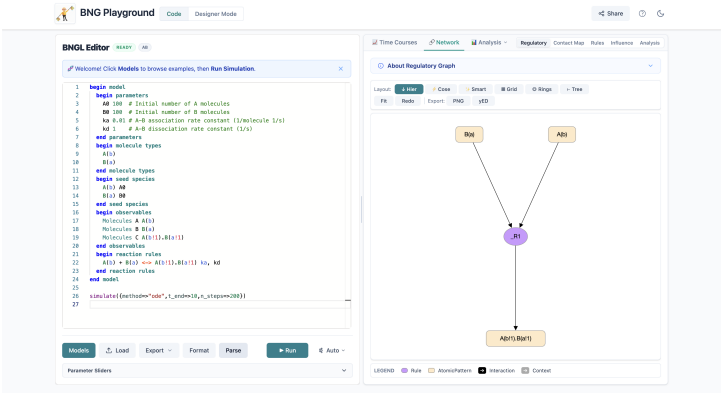



BNGL editor (left) and the regulatory graph of a binding rule (right)

- No installation: write, simulate, and analyze models

ruleworld.github.io/bngplayground

# BNG Playground: BioNetGen in your browser

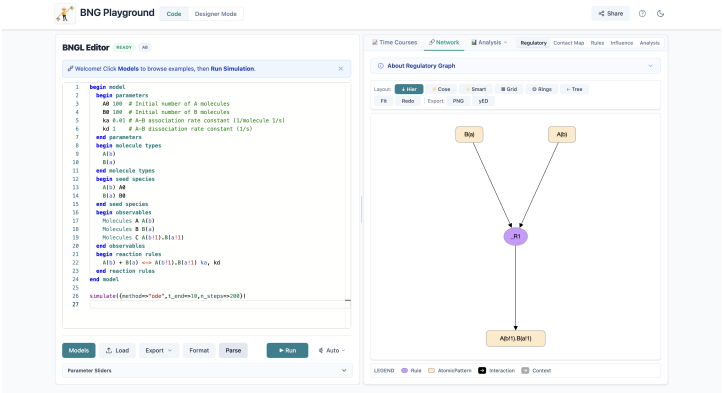



BNGL editor (left) and the regulatory graph of a binding rule (right)

- No installation: write, simulate, and analyze models
- Built-in examples

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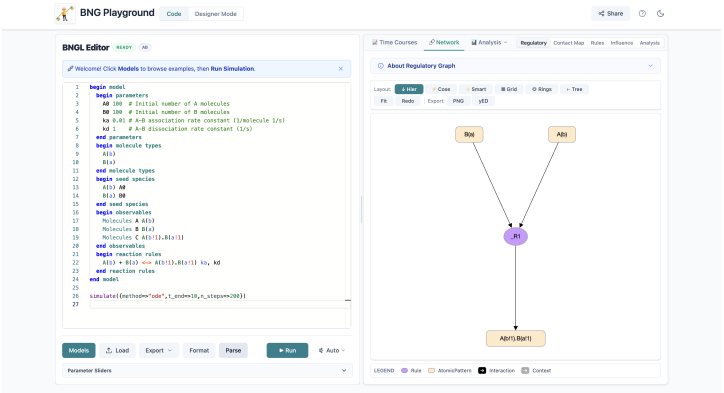



BNGL editor (left) and the regulatory graph of a binding rule (right)

- No installation: write, simulate, and analyze models
- Built-in examples
- Share a model as a link

ruleworld.github.io/bngplayground

# BNG Playground: BioNetGen in your browser




BNGL editor (left) and the regulatory graph of a binding rule (right)

- No installation: write, simulate, and analyze models
- Built-in examples
- Share a model as a link
- Paste in any file from `models/`

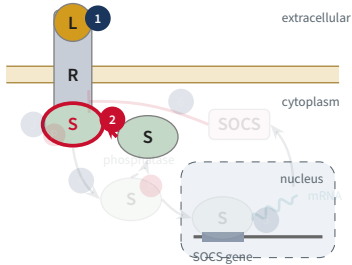
ruleworld.github.io/bngplayground

Part 4

# Building the signaling circuit in BNGL

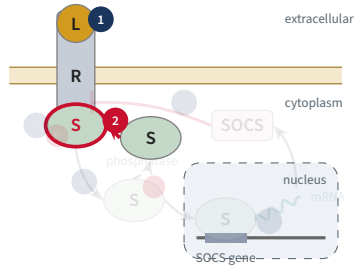
# Step 2: STAT binds the ligand-bound receptor

$$LR + S \rightleftharpoons LRS$$



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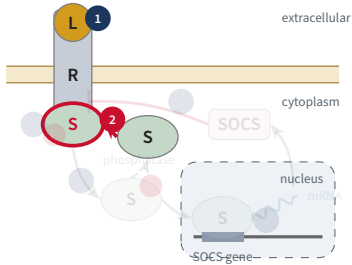

STAT can bind *only* the ligand-bound receptor LR, never free R. This is how ligand binding **enables** STAT recruitment.

# Step 2: STAT binds the ligand-bound receptor

$$LR + S \rightleftharpoons LRS$$

New lines in `stat_step2_recruit.bngl`:

```
begin parameters
  S0     10   # STAT (uM)
  kSon   1    # STAT binding (1/uM/min)
  kSoff  1    # STAT unbinding (1/min)
end parameters
begin molecule types
  S
  LRS
end molecule types
begin species
  S  S0
end species
begin reaction rules
  Sbind: LR + S <-> LRS kSon, kSoff
end reaction rules
```



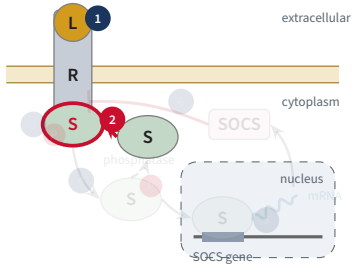

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begin parameters
  S0     10   # STAT (uM)
  kSon   1    # STAT binding (1/uM/min)
  kSoff  1    # STAT unbinding (1/min)
end parameters
begin molecule types
  S
  LRS
end molecule types
begin species
  S  S0
end species
begin reaction rules
  Sbind: LR + S <-> LRS kSon, kSoff
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```



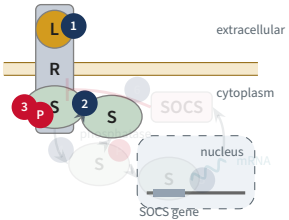

STAT can bind *only* the ligand-bound receptor LR, never free R. This is how ligand binding **enables** STAT recruitment.

Every new species needs a molecule type; species present at $t = 0$ also need an initial amount.

# Step 3: receptor-bound STAT is activated

$$\mathrm{LRS} \xrightarrow{k_{\mathrm{act}}} \mathrm{LRS_p}$$

A receptor-associated kinase (JAK) phosphorylates the docked STAT: a **first-order**, irreversible reaction with $v = k_{\mathrm{act}}[\mathrm{LRS}]$.



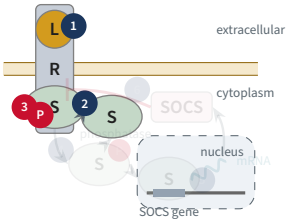
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```
kact  1   # STAT activation (1/min)
LRSp
Sact: LRS -> LRSp kact
```

one new line each in parameters, molecule types, and reaction rules



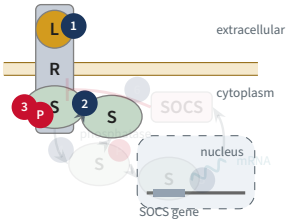
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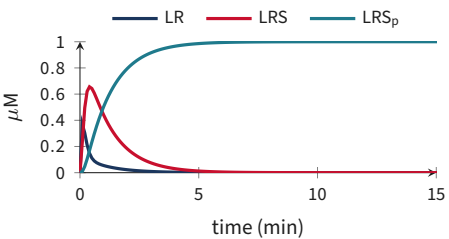
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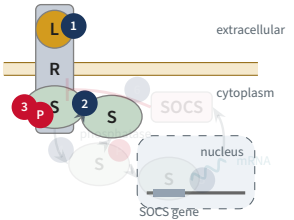
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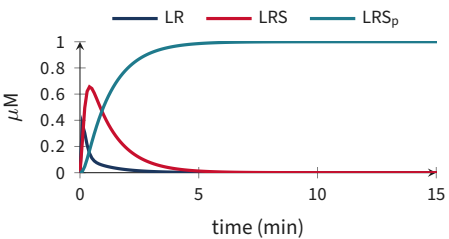
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```
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LRSp
Sact: LRS -> LRSp kact
```

one new line each in parameters, molecule types, and reaction rules

**Problem:** pSTAT never leaves the receptor. Every receptor ends up stuck as $\mathrm{LRS_p}$.

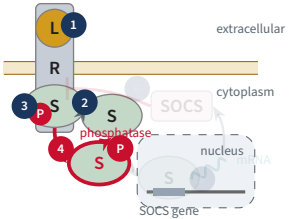


# Step 4: activated STAT dissociates from the receptor

$$LRS_p \xrightarrow{k_{diss}} LR + S_p \qquad S_p \xrightarrow{k_{dephos}} S$$

Releasing pSTAT frees the receptor to activate another STAT. A phosphatase returns free pSTAT to the inactive pool.

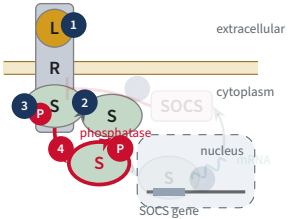


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```
kdiss    1    # pSTAT release (1/min)
kdephos  0.1  # dephosphorylation (1/min)
Sp
Srelease: LRSp -> LR + Sp kdiss
Sdephos:  Sp -> S kdephos
```

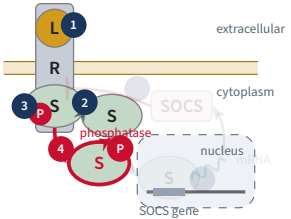


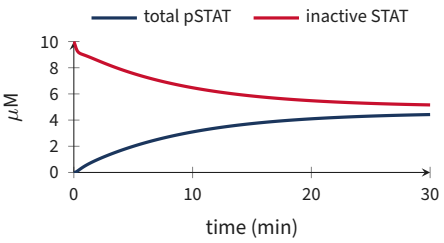
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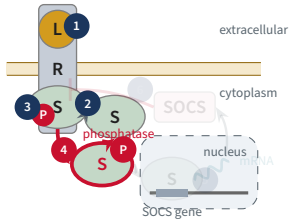
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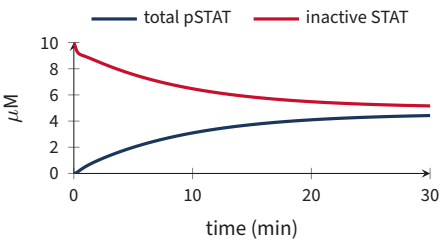
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Releasing pSTAT frees the receptor to activate another STAT. A phosphatase returns free pSTAT to the inactive pool.

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Srelease: LRSp -> LR + Sp kdiss
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```

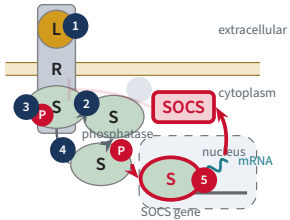
pSTAT now reaches a steady state set by the balance of activation and dephosphorylation.



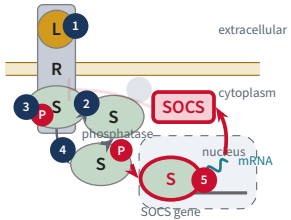
# Step 5: pSTAT induces SOCS expression

```
# pSTAT binds the SOCS promoter
Gbind: Sp + G <-> GSp kGon, kGoff
# Bound promoter is transcribed
Transcribe: GSp -> GSp + mSOCS ktx
# mRNA degradation
mRNAdeg: mSOCS -> 0 kmdeg
# Translation
Translate: mSOCS -> mSOCS + SOCS ktl
# SOCS protein degradation
SOCSdeg: SOCS -> 0 kSOCSdeg
```



# Step 5: pSTAT induces SOCS expression

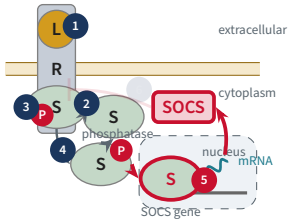
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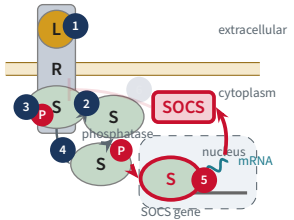


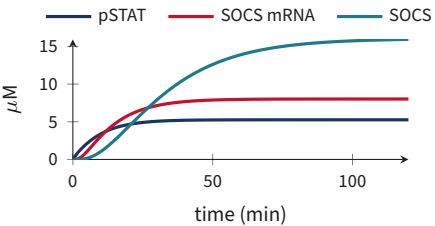

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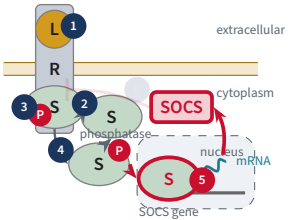



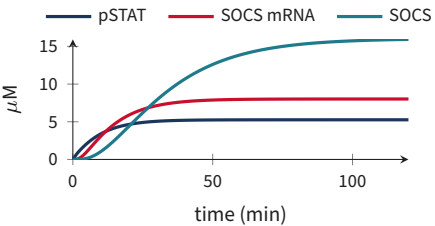
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SOCS accumulates, but it has no target yet, so pSTAT is unaffected.

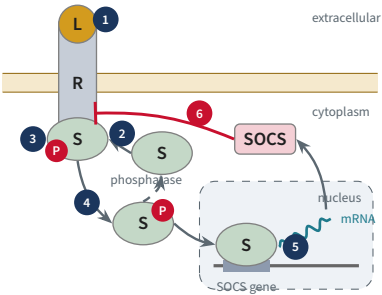


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SOCS competes with STAT for the active receptor. LR·SOCS cannot recruit STAT, so pSTAT production drops.

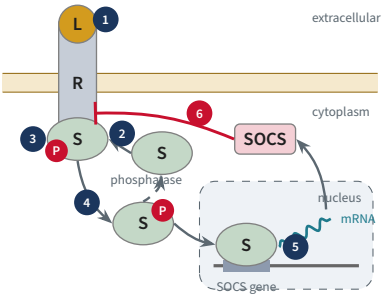


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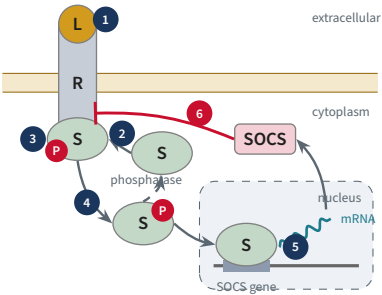
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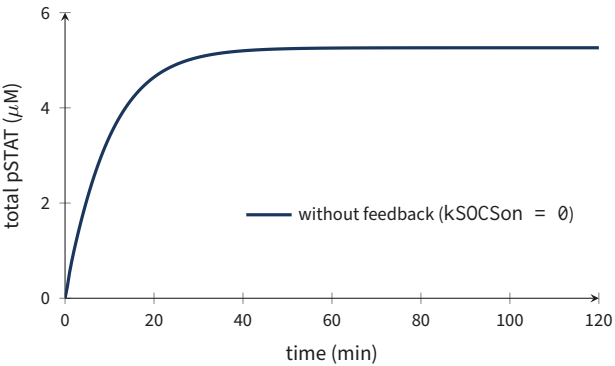
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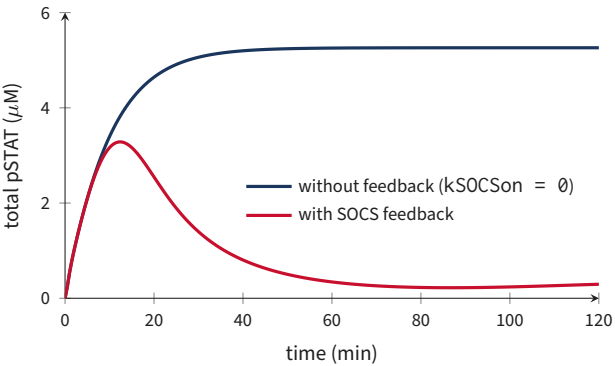
```
kSOCSon   1     # (1/uM/min)
kSOCSoff  0.01  # (1/min)
LRSOCS
SOCSbind: LR + SOCS <-> LRSOCS kSOCSon, kSOCSoff
```



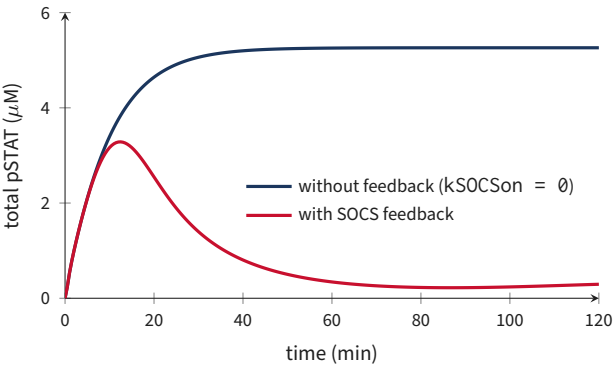
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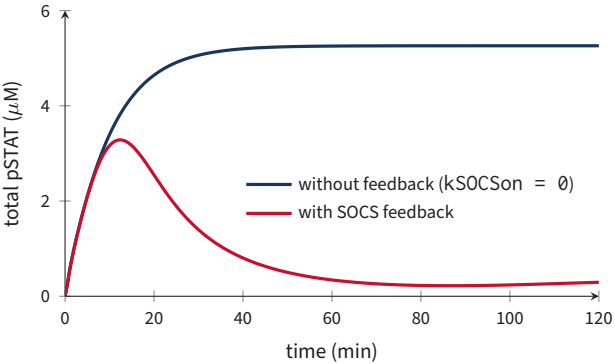


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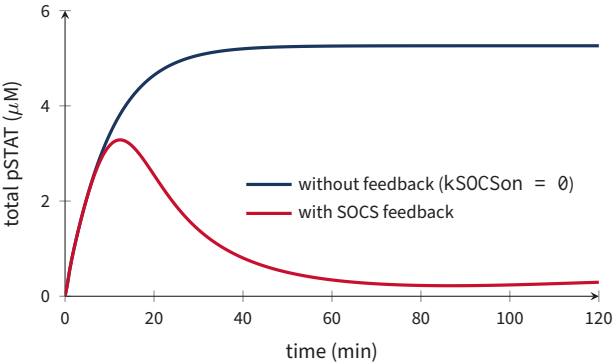

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- The delay comes from transcription and translation.

Both curves come from `stat_step6_feedback.bngl`, which runs the model twice using `setParameter`.

# The complete network

| | reaction | rate |
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| 2 | $LR + S \rightleftharpoons LRS$ | $k_{Son}, k_{Soff}$ |
| 3 | $LRS \rightarrow LRS_p$ | $k_{act}$ |
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- Every species had to be named by hand: L, LR, LRS, $LRS_p$, LR·SOCS, …

Part 5

# Looking ahead

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**Module 2: represent molecules with internal structure (binding sites and states) and write rules that generate the network automatically.**

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We return to stochastic simulation in a later module.

# End-of-class exercises

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No local setup yet? Paste any model from `models/` into BNG Playground (`ruleworld.github.io/bngplayground`) and run it in your browser.

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- Building the cytokine–STAT–SOCS circuit one step at a time shows how each mechanism contributes; negative feedback produces an adaptive pulse of pSTAT.
- Writing every complex as a separate species quickly becomes unmanageable. This motivates rule-based modeling, the subject of Module 2.